Yuming Fu

Curriculum vitae

Leiden Observatory
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Employment

01-06-2023 – **Postdoctoral Researcher**, *Leiden Observatory, Leiden University*, The Netherlands
- Research areas: Quasar and galaxy evolution, observational cosmology, data mining and machine learning in astronomy, astronomical data processing
- Advisors: Dr. R.J. Bouwens (Leiden), and Prof. Dr. Karina Caputi (Groningen)

Sep. 2021 – May. 2023 **CSST Postdoctoral Research Fellow**, *Kavli Institute for Astronomy and Astrophysics, Peking University*, Beijing, China
- Advisor: Prof. Xue-Bing Wu

Academic service

01-06-2023 – **Active member**, *Galaxy/AGN Evolution Science Working Group; NIR Processing Function*, Euclid Consortium
- References: Prof. Lingyu Wang (Groningen), and Dr. Gianluca Polenta (ASI)

01-12-2025 – **Member**, *ESA Euclid Data Space User Group*, `https://euclid.dataspace.esa.int/eds-user-group`
- Reference: Dr. Valeria Pettorino (ESA)

2021– **Peer reviewer**, *A&A, ApJ, ApJS, ATI, Nat Astron, RAA*

Education

Sep. 2016 – Jul. 2021 **Ph.D. in Astrophysics**, *Department of Astronomy, Peking University*, Beijing
- Supervisor: Prof. Xue-Bing Wu
- Thesis: A Survey for Quasars behind the Galactic Plane
- Outstanding Graduate of Beijing Higher Education Institutions (2021)

Apr. 2019 – Aug. 2019 **Visiting Ph.D. student**, *Leiden Observatory*, Leiden
- Supervisor: Prof. Dr. Anthony G. A. Brown

Sep. 2012 – Jul. 2016 **Bachelor of Science (Geophysics)**, *School of Geophysics and Information Technology, China University of Geosciences*, Beijing
- Supervisor: Prof. Guoming Jiang
- Thesis: High-precision Research on the Velocity Structure of Double Seismic Zone beneath Tohoku Region in Japan
- Outstanding Graduate of Beijing Higher Education Institutions (2016)

Teaching & supervision

2024– **Co-supervisor**, *PhD student T. Pan*, with Prof. Rottgering and Dr. van Weeren, Leiden Observatory

Jul. 2022 **Lecturer**, *Lesson: Decomposition of the one-dimensional spectra of quasars and galaxies*, CSST Summer School on Galaxy Sciences, Peking University

Sep. 2018 – Jul. 2022 **Class advisor**, *The Undergraduate Class of 2018*, Department of Astronomy, School of Physics, Peking University

Sep. 2019 – Jun. 2023 **Co-supervisor of seven students**, *Astronomical BSc Thesis Projects*, Peking University

Sep. 2019 – Jan. 2020 **Teaching assistant**, *Course: Observational Experiments for Astrophysics (Optical)*, Peking University

High-Performance Computing & Infrastructure

2017 – 2023 **Computational infrastructure engineering**, *Design, configuration, and maintenance of a system with dual Intel Xeon E5-2699 v4 CPUs (44 cores, 256 GB RAM, 150 TB storage), delivered an estimated theoretical peak performance of* ***1.55 TFLOPS***, Peking University

2022 – 2025 **Research computing user**, *Experienced in using Peking University's supercomputing clusters, Weiming-1 (542 TFLOPS) and Weiming-2 (2.71 PFLOPS), for large-scale astronomical machine learning tasks*, Peking University

Grants

2022 **Candidate Selections for the CSST Quasar Survey with Multi-band Photometry and Slitless Spectroscopy**, *General Program of China Postdoctoral Science Foundation, No. 2022M720266*, 80,000 CNY, PI

2022 – 2026 **A Survey of Quasars behind the Galactic Plane**, *NSFC Key Grant, No. 12133001*, 3.10 million CNY, Co-I

First-author papers

[5] **Euclid Collaboration: Fu, Y.**, Bouwens, R., Caputi, K. I., Vergani, D., Scialpi, M., et al. "Euclid Quick Data Release (Q1). XXXVIII. Euclid spectroscopy of quasars. 1. Identification and redshift determination of 3500 bright quasars". 2026, *A&A*, 711, A285. DOI: `10.1051/0004-6361/202558490`.

[4] **Fu, Y.**, Wu, X.-B., Bouwens, R. J., Caputi, K. I., Pang, Y., et al. "The CatSouth Quasar Candidate Catalog for the Southern Sky and a Unified All-Sky Catalog Based on Gaia DR3". 2025, *ApJS*, 279, 2, 54. DOI: `10.3847/1538-4365/ade999`.

[3] **Fu, Y.**, Wu, X.-B., Li, Y., Pang, Y., Joshi, R., et al. "CatNorth: An Improved Gaia DR3 Quasar Candidate Catalog with Pan-STARRS1 and CatWISE". 2024, *ApJS*, 271, 2, 54. DOI: `10.3847/1538-4365/ad2ae6`.

[2] **Fu, Y.**, Wu, X.-B., Jiang, L., Zhang, Y., Huo, Z.-Y., et al. "Finding Quasars behind the Galactic Plane. II. Spectroscopic Identifications of 204 Quasars at $|b| < 20°$". 2022, *ApJS*, 261, 2, 32. DOI: `10.3847/1538-4365/ac7f3e`.

[1] **Fu, Y.**, Wu, X.-B., Yang, Q., Brown, A. G. A., Feng, X., et al. "Finding Quasars behind the Galactic Plane. I. Candidate Selections with Transfer Learning". 2021, *ApJS*, 254, 1, 6. DOI: `10.3847/1538-4365/abe85e`.

Second-, third-, and corresponding* author papers

[11] Feng, X., Wu, X.-B., **Fu, Y.**, Pang, Y., Zhu, R., et al. "Far-Infrared Star Formation Rates of Quasar Host Galaxies from Multiwavelength Spectral Energy Distribution Decomposition". 2026, *Universe*, 12, 7, 213. DOI: `10.3390/universe12070213`.

[10] Feng, X., Wu, X.-B., **Fu, Y.**, Pang, Y., Zhu, R., et al. "Testing [O ii] λ3727 as a Star Formation Rate Tracer in Quasar Host Galaxies". 2026, *Universe*, 12, 7, 214. DOI: `10.3390/universe12070214`.

[9] Pan, T., **Fu*, Y.**, Rottgering, H. J. A., de Jong, J. M. G. H. J., Hardcastle, M. J., et al. "Probing the environments of FRI and FRII radio galaxies in LoTSS DR2 with galaxy clusters". 2026, *A&A*, 710, A326. DOI: 10.1051/0004-6361/202557097.

[8] Qin, J., Wu, X.-B., **Fu, Y.**, Xu, H., Pang, Y., et al. "Exploring the S_8 Tension: Insights from the CatNorth 1.5 Million Quasar Candidates". 2026, *ApJ*, 1001, 1, 121. DOI: 10.3847/1538-4357/ae4ee5.

[7] Choi, Y., **Fu, Y.**, Im, M., Wu, X.-B., Onken, C. A., et al. "AllBRICQS: The Discovery of Luminous Quasars in the Northern Hemisphere". 2025, *ApJS*, 280, 2, 73. DOI: 10.3847/1538-4365/adf8ed.

[6] He, Z., **Fu, Y.**, Wu, X., and He, L. "Identifying FeLoBAL Quasars in SDSS DR7Q with the Convolutional Neural Network". 2025, *ChA&A*, 49, 4, 756–772. DOI: 10.1016/j.chinastron.2025.11.005.

[5] Huo, Z.-Y., **Fu, Y.**, Huang, Y., Yuan, H., Wu, X.-B., et al. "Finding Quasars behind the Galactic Plane. III. Spectroscopic Identifications of $\sim$1300 New Quasars at $|b| \leq 20°$ from LAMOST DR10". 2025, *ApJS*, 278, 1, 6. DOI: 10.3847/1538-4365/adba52.

[4] Pan, T., **Fu, Y.**, Rottgering, H. J. A., van Weeren, R. J., Drake, A. B., et al. "The environments of radio galaxies and quasars in LoTSS data release 2". 2025, *A&A*, 695, A69. DOI: 10.1051/0004-6361/202453154.

[3] Pang, Y., Wu, X.-B., **Fu, Y.**, et al. "A Pilot Study for the CSST Slitless Spectroscopic Quasar Survey Based on Mock Data". 2025, *ApJ*, 980, 223. DOI: 10.3847/1538-4357/adabdc.

[2] Pang, Y., Wu, X.-B., **Fu, Y.**, Zhu, R., Ji, T., et al. "Quasar identifications from the slitless spectra: a test from 3D-HST". 2025, *MNRAS*, 540, 3, 2216–2237. DOI: 10.1093/mnras/staf849.

[1] Jin, J.-J., Wu, X.-B., **Fu, Y.**, et al. "The Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST) Quasar Survey: Quasar Properties from Data Releases 6 to 9". 2023, *ApJS*, 265, 1, 25. DOI: 10.3847/1538-4365/acaf89.

Contributed Euclid papers (selected)

[10] Belladitta, S., Decarli, R., Bañados, E., et al. "Euclid: A UV-faint quasar in a highly luminous star-forming host galaxy at $z \approx 7.7$". 2026, *A&A, accepted*, arXiv:2607.03430.

[9] Euclid Collaboration: Matamoro Zatarain, T., Fotopoulou, S., Ricci, F., et al. "Euclid Quick Data Release (Q1). XX. The active galaxies of Euclid". 2026, *A&A*, 711, A20. DOI: 10.1051/0004-6361/202554619.

[8] Euclid Collaboration: Polenta, G., Frailis, M., Alavi, A., et al. "Euclid Quick Data Release (Q1). III. NIR processing and data products". 2026, *A&A*, 711, A3. DOI: 10.1051/0004-6361/202554657.

[7] Euclid Collaboration: Siudek, M., Huertas-Company, M., Smith, M. J., et al. "Euclid Quick Data Release (Q1). XIII. Exploring galaxy properties with a multi-modal foundation model". 2026, *A&A*, 711, A13. DOI: 10.1051/0004-6361/202554611.

[6] Euclid Collaboration: Tarsitano, F., Fotopoulou, S., Banerji, M., et al. "Euclid Quick Data Release (Q1). XIX. First study of red quasars selection". 2026, *A&A*, 711, A19. DOI: 10.1051/0004-6361/202554591.

[5] Tumborang, A. A., Caputi, K. I., Rinaldi, P., et al. "Euclid: Scaled-up little red dots and other sources with v-shaped spectral energy distributions at z>4". 2026, *A&A, submitted*, arXiv:2604.16178.

[4] Yang, D., Hennawi, J. F., Guarneri, F., et al. "Euclid: Discovery of 31 new quasars at $6.6 < z < 7.8$". 2026, *A&A, accepted*, arXiv:2607.03432.

[3] Euclid Collaboration: Mellier, Y., Abdurro'uf, Acevedo Barroso, J., et al. "Euclid - I. Overview of the Euclid mission". 2025, *A&A*, 697, A1. DOI: `10.1051/0004-6361/202450810`.

[2] Euclid Collaboration: Navarro-Carrera, R., Caputi, K. I., McPartland, C. J. R., et al. "Euclid Quick Data Release (Q1): Identification of massive galaxy candidates at the end of the Epoch of Reionisation". 2025, *A&A, submitted*, arXiv:2511.11943. DOI: `10.48550/arXiv.2511.11943`.

[1] Ulivi, L., Mannucci, F., Scialpi, M., et al. "Euclid: A machine-learning search for dual and lensed AGN at sub-arcsec separations". 2025, *A&A, submitted*, arXiv:2508.19494. DOI: `10.48550/arXiv.2508.19494`.

Other co-authored papers

[18] Lyu, B., Wu, X.-B., Jin, J.-J., **Fu, Y.**, Pang, Y., et al. "The Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST) Quasar Survey: Quasar Properties from Data Releases 10 to 12". 2026, *ApJS*, 282, 2, 72. DOI: `10.3847/1538-4365/ae2b6e`.

[17] Shi, J.-H. et al. "Unveiling Quasars in Gaia DR3: Multimodal Classification and Redshift Estimation of BP/RP Sources with Deep Learning". 2026, *ApJS*, 283, 1, 38. DOI: `10.3847/1538-4365/ae4003`.

[16] Wang, H. et al. "Systematic Analysis of Changing-Look Active Galactic Nucleus Variability Using ZTF Light Curves". 2026, *ApJ*, 997, 1, 100. DOI: `10.3847/1538-4357/ae1fdc`.

[15] Wang, S. et al. "A Rare Eddington-limited, Heavily Obscured Low-mass Active Galactic Nucleus Likely Triggered by a Galaxy Merger". 2026, *ApJ*, 1003, 2, 172. DOI: `10.3847/1538-4357/ae6779`.

[14] Wu, D., He, Z., Li, N., Cui, S., **Fu, Y.**, et al. "CURLING: III. Identifying candidate wide-separation gravitationally lensed quasars from the CatNorth catalogue". 2026, *A&A*, 710, A180. DOI: `10.1051/0004-6361/202558223`.

[13] Zhang, X., Ai, Y., Zhang, Y., **Fu, Y.**, Wu, X.-B., et al. "Finding Quasars behind the Galactic Plane. IV. Candidate Selection from Chandra with Random Forest". 2026, *ApJ*, 1002, 1, 76. DOI: `10.3847/1538-4357/ae5c03`.

[12] Zhu, R., Wu, X.-B., Pang, Y., and **Fu, Y.** "QHSC: The Quasar Candidate Catalog for the Hyper Suprime-Cam Subaru Strategic Program". 2026, *ApJS*, 282, 2, 38. DOI: `10.3847/1538-4365/ae2099`.

[11] Lyu, B. et al. "The changing-look AGN SDSS J101152.98+544206.4 is returning to a type I state". 2025, *A&A*, 693, A173. DOI: `10.1051/0004-6361/202451699`.

[10] Wang, H., Ai, Y., Zhang, Y., **Fu, Y.**, Wen, W., et al. "Highly Variable Quasar Candidates Selected from 4XMM-DR13 with Machine Learning". 2025, *ApJ*, 985, 1, 23. DOI: `10.3847/1538-4357/adc7b8`.

[9] Yang, Q. et al. "Galaxies Lighting Up: Discovery of Seventy New Turn-on Changing-look Active Galactic Nuclei". 2025, *ApJ*, 980, 1, 91. DOI: `10.3847/1538-4357/ad94ed`.

[8] Liu, C. et al. "Forecasting supernova observations with the CSST: I. Photometric samples". 2024, *Science China Physics, Mechanics, and Astronomy*, 67, 11, 119512. DOI: 10.1007/s11433-024-2456-x.

[7] Liu, S., Luo, A.-L., Zheng, Z., Zhang, W., **Fu, Y.-M.**, et al. "The origin of the X-ray luminosity of the green pea galaxies: X-ray binaries or active galactic nuclei?" 2024, *A&A*, 689, A170. DOI: 10.1051/0004-6361/202449406.

[6] Ma, Q., Wen, Y., Wu, X.-B., Gu, H., and **Fu, Y.** "Hα Time Delays of Active Galactic Nuclei from the Zwicky Transient Facility Broadband Photometry". 2024, *ApJ*, 966, 1, 5. DOI: 10.3847/1538-4357/ad34d6.

[5] Ma, Q., Wu, X.-B., Gu, H., Wen, Y., and **Fu, Y.** "The Hα Broadband Photometric Reverberation Mapping of Four Seyfert 1 Galaxies". 2023, *ApJ*, 949, 1, 22. DOI: 10.3847/1538-4357/acc4c1.

[4] Li, J.-T., Wang, F., Yang, J., Zhang, Y., **Fu, Y.**, et al. "Chandra Detection of Three X-Ray Bright Quasars at $z > 5$". 2021, *ApJ*, 906, 2, 135. DOI: 10.3847/1538-4357/abc750.

[3] Yao, S. et al. "The Large Sky Area Multi-object Fiber Spectroscopic Telescope (LAMOST) Quasar Survey: The Fourth and Fifth Data Releases". 2019, *ApJS*, 240, 1, 6. DOI: 10.3847/1538-4365/aaef88.

[2] Yang, Q. et al. "Discovery of 21 New Changing-look AGNs in the Northern Sky". 2018, *ApJ*, 862, 2, 109. DOI: 10.3847/1538-4357/aaca3a.

[1] Yang, Q. et al. "Quasar Photometric Redshifts and Candidate Selection: A New Algorithm Based on Optical and Mid-infrared Photometric Data". 2017, *AJ*, 154, 6, 269. DOI: 10.3847/1538-3881/aa943c.

Software

[5] **Fu, Y.** *euclidkit: a comprehensive Python package for Euclid archival data analysis*. Version v0.2.4. July 2026. URL: https://github.com/rudolffu/euclidkit.

[4] **Fu, Y.** *qsospec: a fast and robust Python package for fitting quasar spectra*. Version v0.1.1. June 2026. URL: https://github.com/rudolffu/qsospec.

[3] **Fu, Y.** *specbox: a simple tool to manipulate and visualize UV/optical/NIR spectra for astronomical research*. Version v1.0.0. Feb. 2026. DOI: 10.5281/zenodo.18642758.

[2] **Fu, Y.** *QSOFITMORE: a python package for fitting UV-optical spectra of quasars*. Version v1.2.2. June 2025. DOI: 10.5281/zenodo.15571037.

[1] **Fu, Y.** *PyFOSC: a pipeline toolbox for BFOSC/YFOSC long-slit spectroscopy data reduction*. Version v1.1.0. Apr. 2024. DOI: 10.5281/zenodo.10967240.

Conference Talks

Jun. 2026 **Euclid quasar spectroscopy from Q1 to DR1**, *European Astronomy Society (EAS) 2026 Annual Meeting (Euclid: the first 2000 square degrees of the cosmos)*, Lausanne, Switzerland

Dec. 2025 **Identification of 3500 Bright QSOs with Euclid Q1 Spectroscopy (invited)**, *Working with Euclid Q1: From Data Access to Science*, Madrid, Spain

Mar. 2025 **4000 Bright Quasar Candidates from Gaia DR3 in Euclid Q1**, *ESLAB# 56 and 2025 Euclid Consortium Meeting*, Leiden, Netherlands

Jan. 2025 **Highly Complete Bright Quasar Sample from Gaia DR3 in Euclid Q1: Insights from the CatNorth+CatSouth Catalogues**, *EUCLID – From Q1 to DR1. Joint Workshop of Euclid Local Universe, Galaxy and AGN Evolution, and Primeval Universe Science Working Groups*, Tenerife, Spain

Jun. 2024 **Identifying the Rare FeLoBAL Quasars: Generative and Active Learning with Limited Data**, *2024 Euclid Consortium Meeting*, Rome, Italy

Feb. 2024 **Synergies Between Euclid and CSST in Galaxy and AGN Surveys: Indications from the First Euclid Data**, *2024 Euclid Galaxy and AGN Evolution Science Working Group Workshop*, Bologna, Italy

Dec. 2021 **Finding Iron Low-ionization Broad Absorption Line Quasars with SDSS**, *2020 Annual Meeting of Chinese Astronomical Society*, Nanchong, China

May. 2021 **A Survey for Quasars behind the Galactic Plane**, *The 23rd CAS Guoshoujing Symposium on Galaxies and Cosmology*, Hangzhou, China
The Best Oral Presentation

Apr. 2019 **Uncovering Hidden Quasars behind the Galactic Plane with AllWISE, Pan–STARRS and Gaia**, *ESLAB# 53: The Gaia Universe*, Noordwijk, Netherlands

Observations

2017 – 2022 **The 2.16-meter Telescope**, *Xinglong Observatory, National Astronomical Observatories, Chinese Academy of Sciences (NAOC)*, PI/Co-I, on-site/remote
Allocated time: > 350 hours

2017 – 2022 **The 2.4-meter Telescope**, *Lijiang Observatory, Yunnan Observatories (YNAO), Chinese Academy of Sciences*, PI/Co-I, on-site/remote
Allocated time: > 180 hours

2019 – 2021 **The McGraw-Hill 1.3-m Telescope**, *MDM Observatory*, PI/Co-I, remote
Allocated time: > 100 hours

2018 – 2024 **The 200-inch Hale Telescope (P200)**, *Palomar Observatory*, PI/Co-I, remote
Allocated time: > 90 hours

Training/Schools

Jul. 2018 **The 2nd East Asian Workshop on Astrostatistics**, *Purple Mountain Observatory*, Nanjing, China

Nov. 2017 **The Ninth Xinglong Observational Astrophysics Workshop: High Resolution Spectroscopy**, *Xinglong Observatory, NAOC*, Xinglong, China

Sep. 2017 **TIARA Summer School on Astrostatistics and Big Data**, *TIARA, ASIAA*, Tapei, China

Jul. 2017 **LAMOST Users Workshop 2017**, *NAOC/UCAS*, Beijing, China

Aug. 2016 **The Sixth Xinglong Observational Astrophysics Workshop: Time Domain Astronomy**, *Xinglong Observatory, NAOC*, Xinglong, China

Skills

Observation **Photometric and spectroscopic observations. Data reduction with IRAF, Python, etc.**

Coding **Python, MATLAB, Shell, Fortran, C**

Database **ADQL, TOPCAT, MySQL, ClickHouse**

Big Data **Data Mining, Statistical Machine Learning, Deep Learning, LLM**

Outreach

Jan. 2024 **Public Talk: The Beauty of the Space II: the Universe as seen by JWST and Euclid**, *Library of Jianyang City*, Chengdu

Mar. 2022 **Public Talk: Quasars in the Distant Universe**, *AstroSalon# 6 of the Peking University Youth Astronomy Society*, Beijing

Jan. 2022 **Public Talk: The Beauty of the Space**, *Library of Jianyang City*, Chengdu

Jan. 2020 **Public Talk: A Brief History of Black Holes**, *Library of Jianyang City*, Chengdu

Jan. 2019 **Public Talk: The Big Eyes: A Glimpse at the Telescopes**, *Library of Jianyang City*, Chengdu

Feb. 2018 **Public Talk: Astronomy in Everyday Life**, *Hangzhou Arts and Crafts Museum*, Hangzhou

Honors & Awards

2020 **Merit Student**, *Peking University*, Beijing

2018, 2020 **Dedicated Scholarship for Outstanding PhD Students**, *Peking University*, Beijing

2020 **Outstanding Teaching Assistant**, *Peking University*, Beijing

2017, 2019 **Award for Scientific Research**, *Peking University*, Beijing

2017 **The Third Prize of CASC Scholarship**, *Peking University*, Beijing